

# Joan Alcaide-Núñez

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[Website](#) — [LinkedIn](#) — [GitHub](#)

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## Research Experience

### GRB Cosmography with Empirical Energy Relations

June–July 2025

*Osservatorio Astronomico di Brera (OAB-INAF), Italy*

Applied GRB empirical relations to constrain cosmological parameters via  $\chi^2$  minimization. Combined GRB results with binary neutron star simulated populations from Einstein Telescope expectations. Received mentoring on relativistic effects and theoretical foundations of GRBs. Participated in weekly journal clubs and research seminars. [code repository](#) Supervisors: Dr. Giancarlo Ghirlanda, Dr. Om Sharan Salafia

### CAPIBARA Collaboration

2024–Present

*Principal Investigator, Founder*

Student research collaboration developing instrumentation and mission concept for high-energy astronomy CubeSat mission. [project website](#)

### Student Internship at German Space Agency

October 2024

*Institute for Remote Sensing (DLR-IMF), Germany*

Worked with experimental methods and photogrammetry departments on hyperspectral imaging and Landsat thermal data analysis. Conducted field measurements and visited German Space Operations Center (GSOC).

### Type Ia Supernovae Distance Measurements

June–July 2024

*Institute of Space Sciences (ICE-CSIC-IEEC), Spain*

Computed Type Ia SNe distances using infrared and optical photometry from ESO, ATLAS, and ZTF. Fitted cosmological parameters including the Hubble constant. Developed proficiency in astronomical data analysis. Participated in journal clubs and group meetings. Supervisor: Dr. Lluís Galbany

### Cepheid Variable Stars Research

August–December 2023

*Youth & Science Program (year 1)*

Computed Hubble constant using Cepheid data from NASA-IPAC NED database and Konkoly observatory extragalactic cepheid database, recovering  $H_0 = 70.26 \pm 7.1 \text{ kms}^{-1}\text{Mpc}^{-1}$ . First experience with astronomical data analysis, paper reading, and scientific writing. [paper report](#) Supervisors: Dr. Ignasi Pérez-Ràfols, Dr. Laia Casamiquela

## Education

### Ludwig-Maximilians-Universität (LMU) Munich

September 2025 – Present

B.Sc. in Physics

### German School of Barcelona

September 2017 - May 2025

Dual High School Diploma (Germany + Spain), GPA: 1.0/1.0

### Additional Training:

- *Gravity and Black Holes* — Perimeter Institute (GoPhysics!), 2025

- Mathematics, Statistics, and Data Science — Universitat Autònoma de Barcelona (UAB), 2022
- Algorithms and Programming (C++) — Universitat Politècnica de Catalunya (UPC), 2022
- League of Codes (C++) — Harbour Space, 2022-2023

## Awards

**Youth & Science fellowship:** 3-year funded [fellowship](#) of research experiences for high school students. Prestigious scholarship that allowed me to have early first-hand experience in research stays internationally.

**German Physical Society (DPG):** Physics Distinction for Academic Achievement in high school.

**International Astronomy & Astrophysics Competition (IAAC):** Silver Honour and National Award for Spain (2025), Silver Honour (2024), Bronze Honour and National Award Spain (2023). In 2025, top 7% students ( $n > 12000$ ).

**Jugend Forscht:** First Prize Iberia 2025 ("*Cosmological distance measurements with Type Ia Supernovae*"), First Prize and Special Award in Scientific Photography NRW 2025; First Prize Ibera 2024 ("*Recomputing the Hubble constant*"), DLR Internship Award NRW 2024.

## Talks

**Symposium for Space Education Activities (SSEA, contributed)** April 2026  
*Technical University of Munich, TUM*  
 Description.

**De l'aula a l'espai: joves, ions i fotons per entendre l'univers d'altres energies (Contributed)** March 2025  
*CosmoXarxa, CosmoCaixa Barcelona's Science Museum*  
 CosmoXarxa is an initiative created by and for Explainers of the science museum (outreach volunteers). I presented the CAPIBARA project, its objectives and status, both technical, scientific and educative, inviting the public to join the efforts. [slides](#)

**Introduction to Astrophysics (Invited)** May/June 2024  
*German School of Barcelona*  
 Double session (3h) about a naive introduction to astronomy and astrophysics for 11th graders, covered some cosmology (Big Bang theory), star evolution, galaxy morphology and massive bodies (neutron stars and black holes as extreme cases of star evolution). [slides](#)

**L'Exploració Espacial (Invited)** October 2023  
*Escola Canigó*  
 Presentation to pre-school students about human space travel and solar system exploration. Focused on hands-on learning with models and didactic, entertaining content. [slides](#)

**Explainers at CosmoCaixa Museum** 2022-2023  
 Volunteered as explainer at Barcelona's Science Museum explaining demonstrations to visitors on weekends. Participated in events such as Explainers during Winter, Explainers at Barcelona's City Science Fair and CosmoXarxa.

## Skills

**Python:** numpy, scipy, pandas, matplotlib, astropy

**Tools:** Git, GitHub, L<sup>A</sup>T<sub>E</sub>X, Markdown, Visual Studio, NeoVim